
Appetizer: using probability to cover a geometric set

We begin our study of high-dimensional probability with an elegant argument that showcases the usefulness of probabilistic reasoning in geometry.

Recall that a *convex combination* of points $z_1, \dots, z_m \in \mathbb{R}^n$ is a linear combination with coefficients that are non-negative and sum to 1, i.e. it is a sum of the form

$$\sum_{i=1}^m \lambda_i z_i \quad \text{where} \quad \lambda_i \geq 0 \quad \text{and} \quad \sum_{i=1}^m \lambda_i = 1. \quad (0.1)$$

The *convex hull* of a set $T \subset \mathbb{R}^n$ is the set of all convex combinations of all finite collections of points in T :

$$\text{conv}(T) := \{\text{convex combinations of } z_1, \dots, z_m \in T \text{ for } m \in \mathbb{N}\};$$

see Figure 0.1 for illustration.

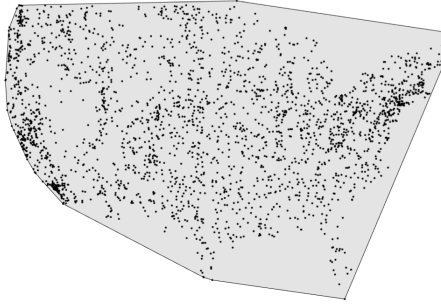


Figure 0.1 The convex hull of a set of points representing major U.S. cities

The number m of elements defining a convex combination in $\mathbb{R}^n$ is not restricted a priori. However, the classical Caratheodory's theorem states that one can always take $m \leq n + 1$.

Theorem 0.0.1 (Caratheodory's theorem). *Every point in the convex hull of a set $T \subset \mathbb{R}^n$ can be expressed as a convex combination of at most $n + 1$ points from T .*

The bound $n + 1$ can not be improved, as it is clearly attained for a simplex T (a set of $n + 1$ points in general position). Suppose, however, that we only want

to *approximate* a point $x \in \text{conv}(T)$ rather than exactly represent it as a convex combination. Can we do it with fewer than $n + 1$ points? We now show that this is possible, and actually the number of required points does not need to depend on the dimension n at all!

Theorem 0.0.2 (Approximate Caratheodory's theorem). *Consider a set $T \subset \mathbb{R}^n$ whose diameter¹ is bounded by 1. Then, for every point $x \in \text{conv}(T)$ and every integer k , one can find points $x_1, \dots, x_k \in T$ such that*

$$\left\| x - \frac{1}{k} \sum_{j=1}^k x_j \right\|_2 \leq \frac{1}{\sqrt{k}}.$$

There are two reasons why this result is surprising. First, the number of points k in convex combinations does not depend on the dimension n . Second, the coefficients of convex combinations can be made all equal. (Note however that repetitions among the points x_i are allowed.)

Proof Our argument is known as the *empirical method* of B. Maurey.

Translating T if necessary, we may assume that not only the diameter but also the *radius* of T is bounded by 1, i.e.

$$\|t\|_2 \leq 1 \quad \text{for all } t \in T. \quad (0.2)$$

Fix a point $x \in \text{conv}(T)$ and express it as a convex combination of some vectors $z_1, \dots, z_m \in T$ as in (0.1). Now, interpret the definition of convex combination (0.1) probabilistically, with λ_i taking the roles of probabilities. Specifically, we can define a random vector Z that takes values z_i with probabilities λ_i :

$$\mathbb{P}\{Z = z_i\} = \lambda_i, \quad i = 1, \dots, m.$$

(This is possible by the fact that the weights λ_i are non-negative and sum to one.) Then

$$\mathbb{E} Z = \sum_{i=1}^m \lambda_i z_i = x.$$

Consider independent copies $Z_1, Z_2, \dots$ of Z . By the the strong law of large numbers,

$$\frac{1}{k} \sum_{j=1}^k Z_j \rightarrow x \quad \text{almost surely as } k \rightarrow \infty.$$

To get a quantitative form of this result, let us compute the variance of $\frac{1}{k} \sum_{j=1}^k Z_j$. (Incidentally, this computation is at the heart of the proof of the weak law of large numbers). We obtain

$$\mathbb{E} \left\| x - \frac{1}{k} \sum_{j=1}^k Z_j \right\|_2^2 = \frac{1}{k^2} \mathbb{E} \left\| \sum_{j=1}^k (Z_j - x) \right\|_2^2 = \frac{1}{k^2} \sum_{j=1}^k \mathbb{E} \|Z_j - x\|_2^2$$

¹ The diameter of T is defined as $\text{diam}(T) = \sup\{\|s - t\|_2 : s, t \in T\}$. We assumed that $\text{diam}(T) = 1$ for simplicity. For a general set T , the bound in the theorem changes to $\text{diam}(T)/\sqrt{k}$. Check this!

since $\mathbb{E}(Z_i - x) = 0$ for each i . The last identity is just a higher dimensional version of the basic fact that the variance of a sum of independent random variables equals the sum of variances; see Exercise 0.0.3 below.

It remains to bound the variances of the terms. We have

$$\begin{aligned}\mathbb{E}\|Z_j - x\|_2^2 &= \mathbb{E}\|Z - \mathbb{E} Z\|_2^2 \\ &= \mathbb{E}\|Z\|_2^2 - \|\mathbb{E} Z\|_2^2 \quad (\text{another variance identity; see Exercise 0.0.3}) \\ &\leq \mathbb{E}\|Z\|_2^2 \leq 1 \quad (\text{since } Z \in T \text{ and using (0.2)}).\end{aligned}$$

We showed that

$$\mathbb{E}\left\|x - \frac{1}{k} \sum_{j=1}^k Z_j\right\|_2^2 \leq \frac{1}{k}.$$

Therefore, there exists a realization of the random variables $Z_1, \dots, Z_k$ such that

$$\left\|x - \frac{1}{k} \sum_{j=1}^k Z_j\right\|_2^2 \leq \frac{1}{k}.$$

Since by construction each Z_j takes values in T , the proof is complete. $\square$

Exercise 0.0.3. ☕☕ Check the following variance identities that we used in the proof of Theorem 0.0.2.

(a) Let $Z_1, \dots, Z_k$ be independent mean zero random vectors in $\mathbb{R}^n$. Show that

$$\mathbb{E}\left\|\sum_{j=1}^k Z_j\right\|_2^2 = \sum_{j=1}^k \mathbb{E}\|Z_j\|_2^2.$$

(b) Let Z be a random vector in $\mathbb{R}^n$. Show that

$$\mathbb{E}\|Z - \mathbb{E} Z\|_2^2 = \mathbb{E}\|Z\|_2^2 - \|\mathbb{E} Z\|_2^2.$$

Let us give one application of Theorem 0.0.2 in computational geometry. Suppose we are given a subset $P \subset \mathbb{R}^n$ and ask to cover it by balls of given radius ε , see Figure 0.2. What is the smallest number of balls needed, and how shall we place them?

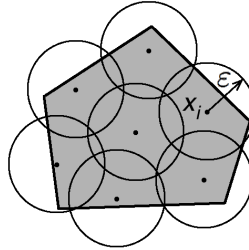


Figure 0.2 The covering problem asks how many balls of radius ε are needed to cover a given set in $\mathbb{R}^n$, and where to place these balls.

Corollary 0.0.4 (Covering polytopes by balls). *Let P be a polytope in $\mathbb{R}^n$ with N vertices and whose diameter is bounded by 1. Then P can be covered by at most $N^{\lceil 1/\varepsilon^2 \rceil}$ Euclidean balls of radii $\varepsilon > 0$.*

Proof Let us define the centers of the balls as follows. Let $k := \lceil 1/\varepsilon^2 \rceil$ and consider the set

$$\mathcal{N} := \left\{ \frac{1}{k} \sum_{j=1}^k x_j : x_j \text{ are vertices of } P \right\}.$$

We claim that the family of ε -balls centered at $\mathcal{N}$ satisfy the conclusion of the corollary. To check this, note that the polytope P is the convex hull of the set of its vertices, which we denote by T . Thus we can apply Theorem 0.0.2 to any point $x \in P = \text{conv}(T)$ and deduce that x is within distance $1/\sqrt{k} \leq \varepsilon$ from some point in $\mathcal{N}$. This shows that the ε -balls centered at $\mathcal{N}$ indeed cover P .

To bound the cardinality of $\mathcal{N}$, note that there are N^k ways to choose k out of N vertices with repetition. Thus $|\mathcal{N}| \leq N^k = N^{\lceil 1/\varepsilon^2 \rceil}$. The proof is complete. $\square$

In this book we learn several other approaches to the covering problem when we relate it to packing (Section 4.2), entropy and coding (Section 4.3) and random processes (Chapters 7–8).

To finish this section, let us show how to slightly improve Corollary 0.0.4.

Exercise 0.0.5 (The sum of binomial coefficients).  Prove the inequalities

$$\left(\frac{n}{m} \right)^m \leq \binom{n}{m} \leq \sum_{k=0}^m \binom{n}{k} \leq \left(\frac{en}{m} \right)^m$$

for all integers $m \in [1, n]$.

Hint: To prove the upper bound, multiply both sides by the quantity $(m/n)^m$, replace this quantity by $(m/n)^k$ in the left side, and use the Binomial Theorem.

Exercise 0.0.6 (Improved covering).  Check that in Corollary 0.0.4,

$$(C + C\varepsilon^2 N)^{\lceil 1/\varepsilon^2 \rceil}$$

suffice. Here C is a suitable absolute constant. (Note that this bound is slightly stronger than $N^{\lceil 1/\varepsilon^2 \rceil}$ for small ε .)

Hint: The number of ways to choose an (unordered) subset of k elements from an N -element set with repetition is $\binom{N+k-1}{k}$. Simplify using Exercise 0.0.5.

0.0.1 Notes

In this section we gave an illustration of the *probabilistic method*, where one employs randomness to construct a useful object. The book [8] presents many illustrations of the probabilistic method, mainly in combinatorics.

The empirical method of B. Maurey we presented in this section was originally proposed in [166]. B. Carl used it to get bounds on covering numbers [49] including those stated in Corollary 0.0.4 and Exercise 0.0.6. The bound in Exercise 0.0.6 is sharp [49, 50].